1)

ex1)

$$T_1 = \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad T_2 = \begin{vmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$T_2^{-1}: R_2^T = 1,\ -R_2^T t_2 = (-3,0,0)^T$

$T_{21} = T_2^{-1} T_1 : R = 1 \cdot 1 = 1$

$t = R_2^T (t_1 - t_2) = (1-3,0,0)^T = (-2,0,0)^T$

$$T_2^{-1} = \begin{vmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad T_{21} = \begin{vmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$p = (0,0,0,1)^T$

$$T_{21} p = \begin{vmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \\ 1 \end{vmatrix} = \begin{vmatrix} -2 \\ 0 \\ 0 \\ 1 \end{vmatrix}$$

Frame 1's origin lies at $(-2,0,0)$ in Frame 2's coordinates

ex2)

$$R_2 = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad R_2^T = \begin{vmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$R_2^T t_2 : \begin{vmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 3 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ -3 \\ 0 \end{vmatrix}$$

2)

a) T_5^4

$P = (L_3, 0) = (1, 0)$

$$T_5^4 = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

T_4^3

$P = (L_2, 0) = (2, 0)$

$$T_4^3 = \begin{vmatrix} C_3 & -S_3 & 2 \\ S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

T_3^2 rotate by θ_2

$P = (L_1, 0) = (2, 0)$

$$T_2^1 = \begin{vmatrix} C_1 & -S_1 & 0 \\ S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$